Containerization 101

with Docker

30/04/2025

Table of content

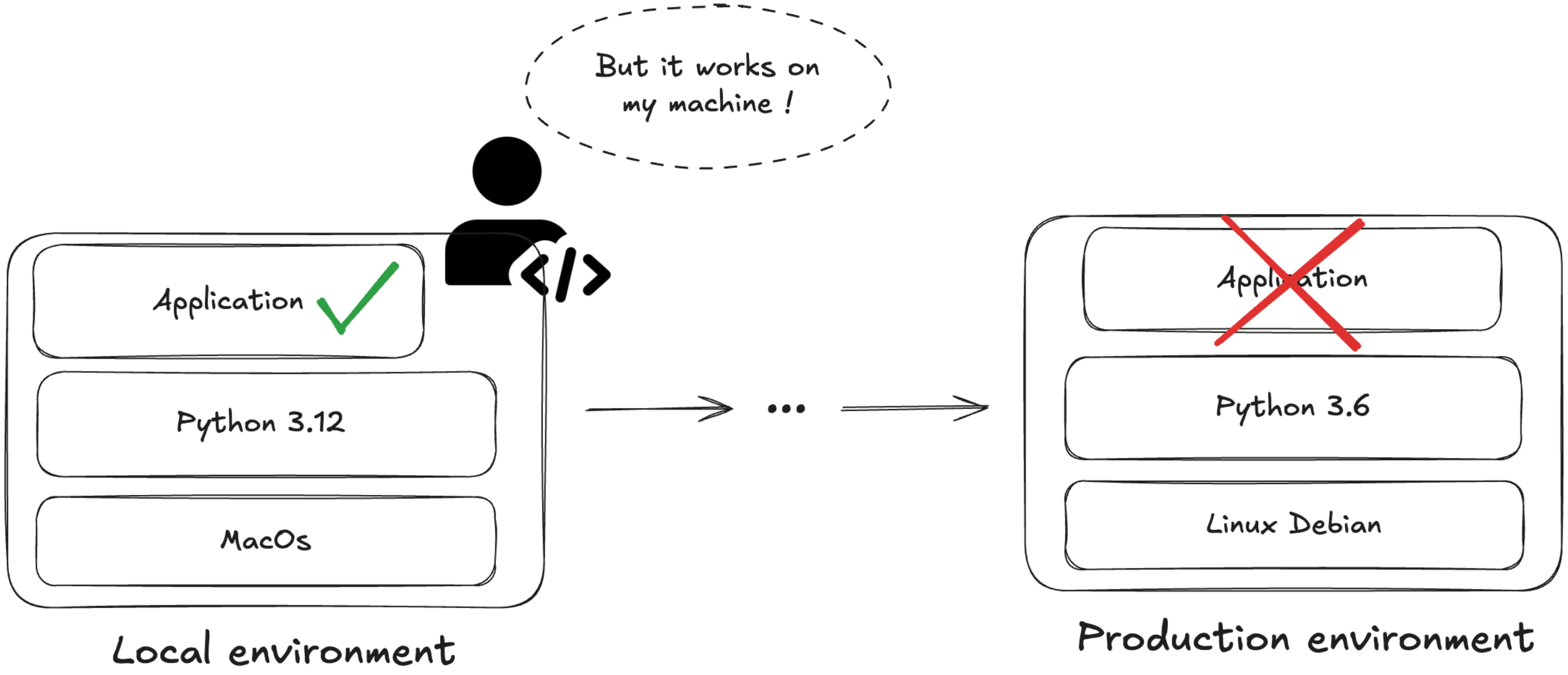
01 What is containerization ?

02 Introduction to Docker

03 Orchestrate Docker containers

01. What is containerization ?

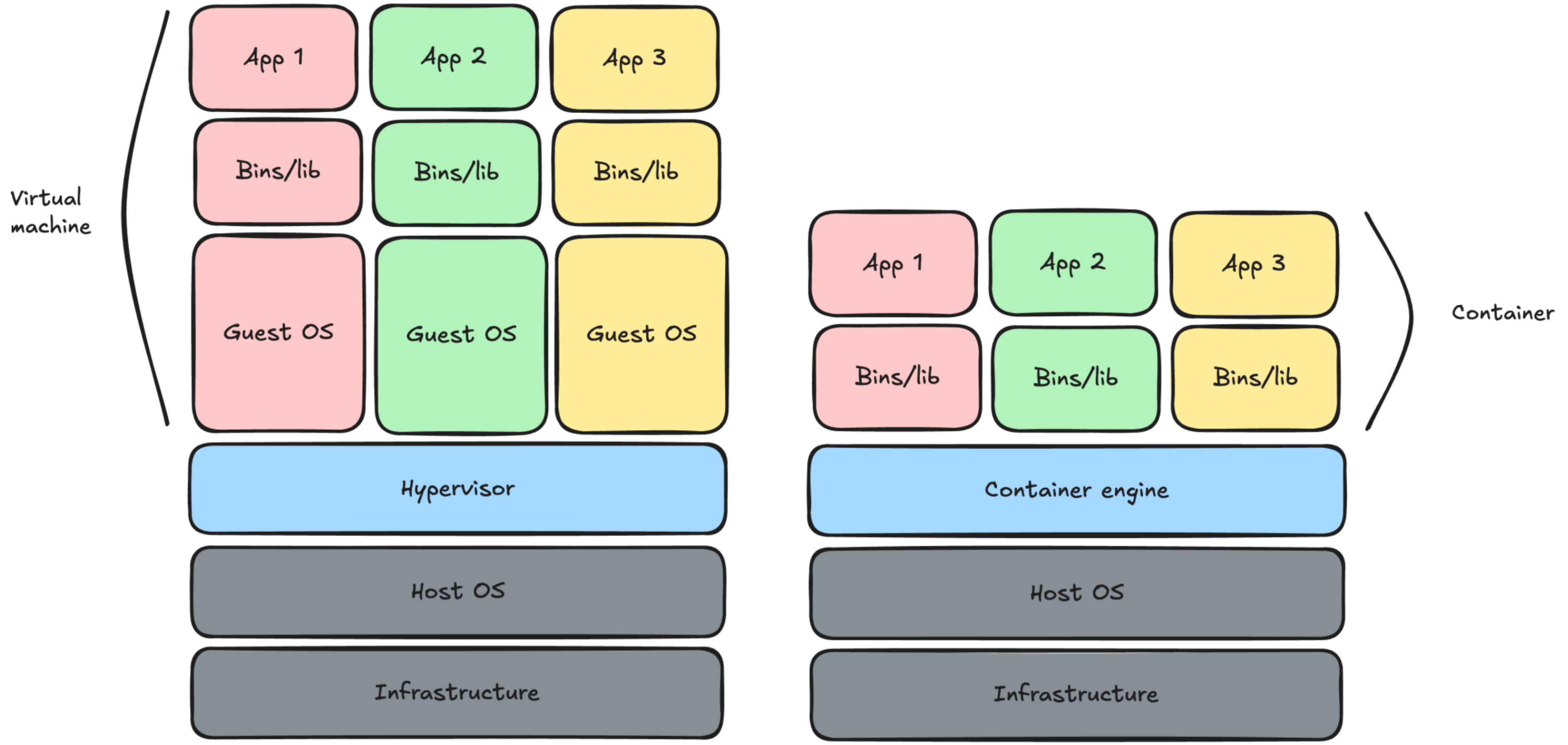
Life without containers



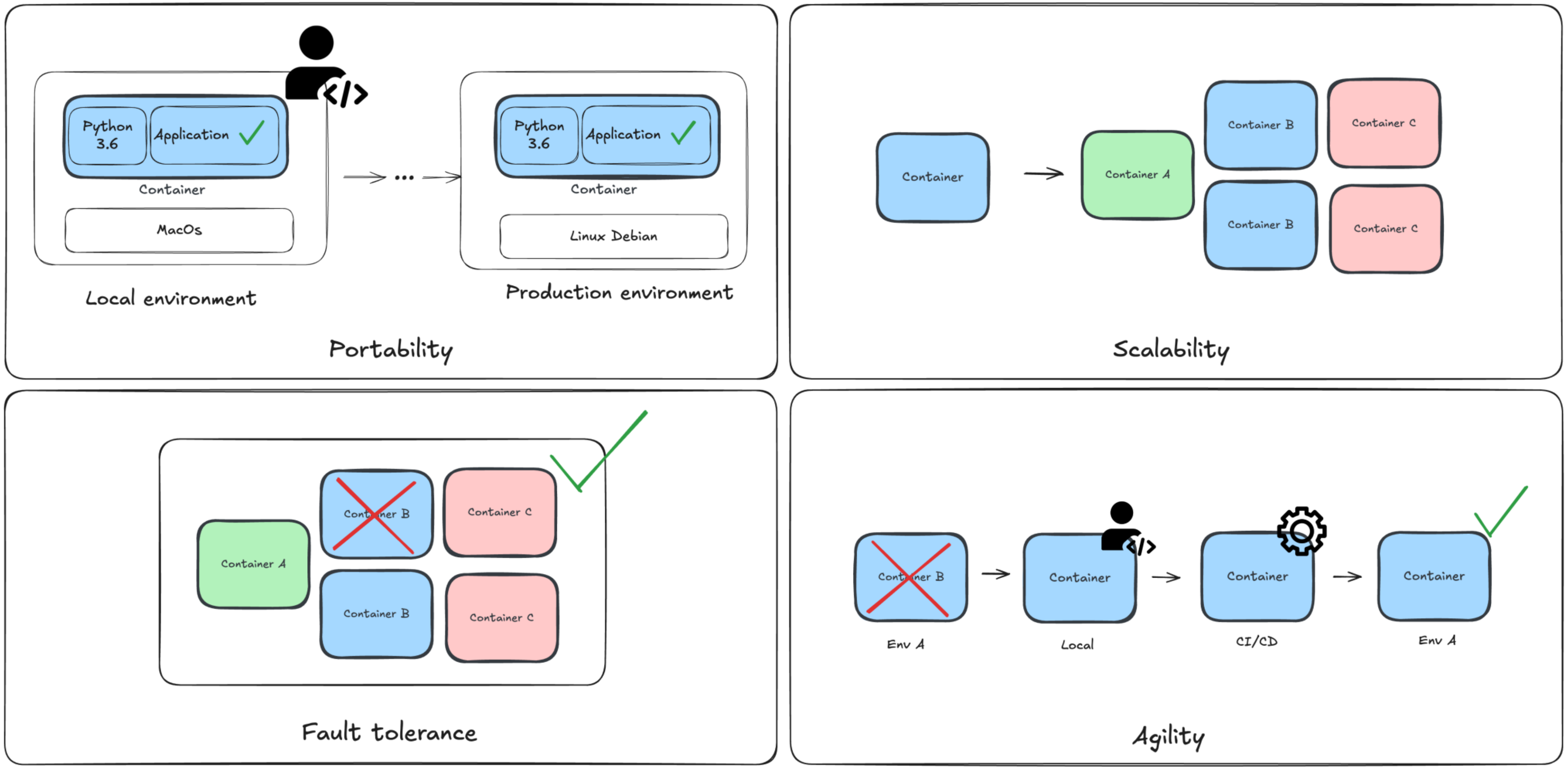
Definition

« Containerization is a software deployment process that bundles an application’s code with all the files and libraries it needs to run on any infrastructure. » AWS

Containers VS Virtual machines



Life with containers



02. Introduction to Docker

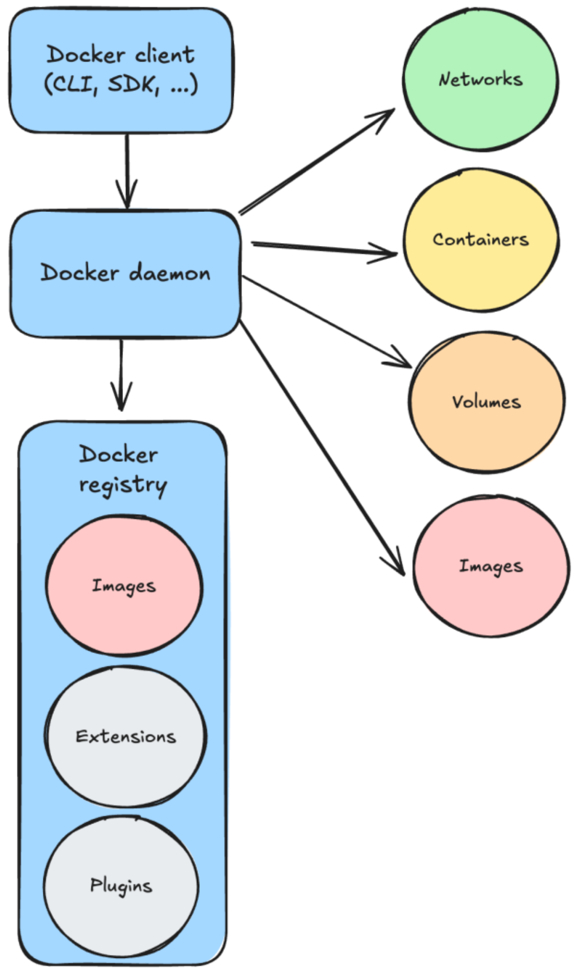
Concepts

Image
Read-only standardized package that includes all of the files, binaries, libraries, and configurations to run a container.

Container
Isolated process with all of the files it needs to run.

Volume
Persistent storage for containers, independent of lifecycle

Network
Communication layer enabling connectivity between containers



Dockerfile

Specify the base image that the build will extend

Copy files from the host and put them into the container image

Run the specified command

```
FROM python:3.12
WORKDIR /usr/local/app

# Install the application dependencies
COPY requirements.txt ./
RUN pip install --no-cache-dir -r requirements.txt

# Copy in the source code
COPY src ./src
EXPOSE 5000

# Setup an app user so the container doesn't run as the root user
RUN useradd app
USER app

CMD ["uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8080"]
```

The path in the image where files will be copied and commands will be executed

Indicates a port the image would like to expose

Sets the default command a container using this image will run

Source: https://docs.docker.com/get-started/docker-concepts/building-images/writing-a-dockerfile/

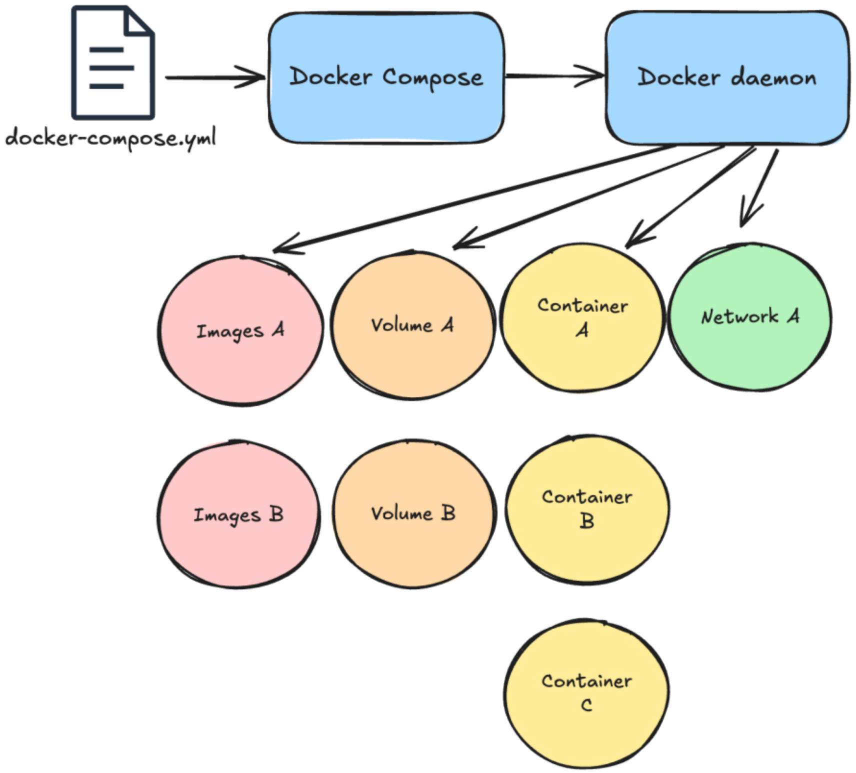
CLI

Build an image	docker build -t <image> .
Run a container	docker run -d -p 8080:80 <image>
Show container logs	docker logs <container>
List running containers	docker ps
Manage local images	docker image ls docker image rm <image>
Download a remote image	docker pull <image>
Manage networks	docker network create <network> docker network ls docker network rm <network>
Manage volumes	docker volume create <volume> docker volume ls docker volume rm <volume>

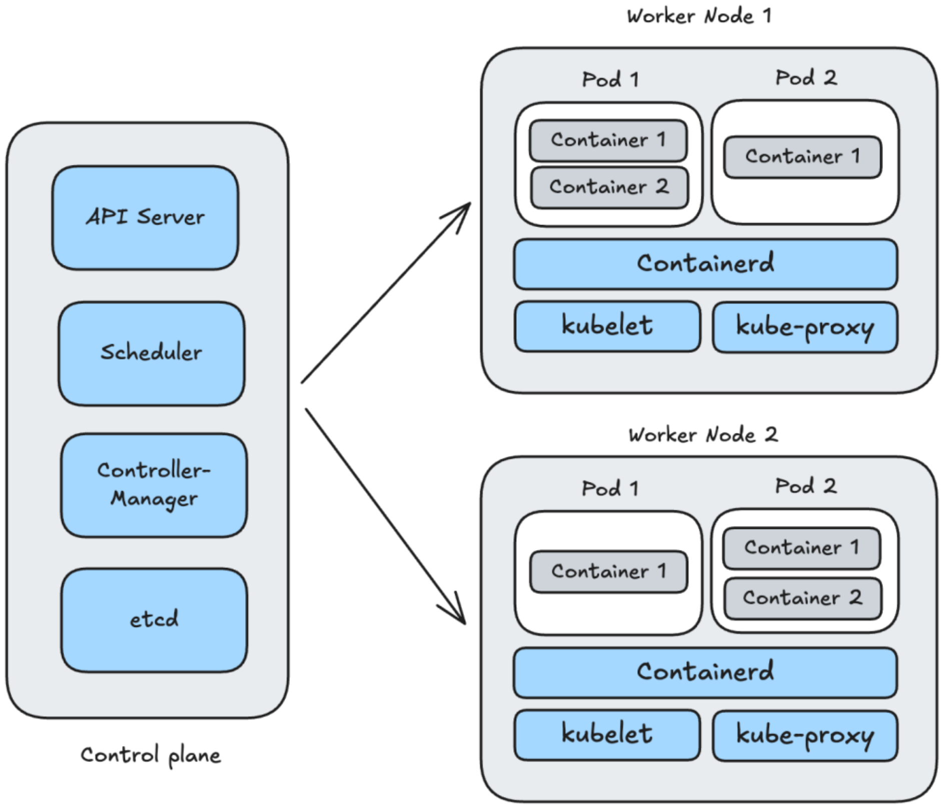
03. Orchestrate Docker containers

Compose & Kubernetes

03. Orchestrate Docker containers



Docker Compose



Kubernetes

Contact

Rayenn HAMROUNI
DevOps consultant